Wood Surface Defect Detection - Methodology Pipeline

End-to-end pipeline from dataset to deployment | YOLOv8n with Class-Aware Augmentation

1. Dataset
4,000 images | 9,211 annot.8 defect classesKaggle dataset

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2. Preprocessing
Resize 640x640YOLO format labelsData cleaning

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3. Data Split
80% Train (3,200)10% Val (400)10% Test (400)

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4. Augmentation
Class-aware oversamplingH/V Flip | +/-10 deg Rot+/-15% Brightness+781 images created

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5. YOLOv8n Training
20 epochs | Batch 16AdamW lr=0.000833Tesla T4 GPU (0.684 hrs)Transfer from COCO

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6. Evaluation
mAP50: 0.639Precision: 0.665Recall: 0.643Best: Dead_Knot 0.852

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7. Deployment
FastAPI + Bootstrap 5Image and Video inferenceReal-time (1.5ms/img)Color-coded detections

Key Performance Summary			
mAP@50	Precision	Recall	Inference Speed
0.639	0.665	0.643	1.5 ms/img